

SimAnyRoom: Making Rooms Simulatable through Evidence-Driven Real-to-Sim

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Abstract—A reconstructed room is not automatically a usable robot simulator: generated objects can have incorrect scale, unstable contact geometry, or appearances that cease to agree after interaction. We present SimAnyRoom, an evidence-driven real-to-sim framework that treats these outputs as proposals requiring verification. Explicit room and object jobs connect specialized tools through persistent artifact contracts. A deterministic controller selects, retries, accepts, rejects, or abstains using construction evidence, while one object identity links metric geometry, collision, and live Gaussian appearance. A complete automatic-discovery controller cohort compares proposal diversity, evidence selection, bounded registration retry, and strict acceptance using shared initial proposals. Evidence selection and retry improve independently matched object geometry at unchanged coverage, with additional computation. Strict acceptance exposes a substantial coverage cost; neither conditional quality nor the acceptance-time physical probe establishes manipulation gains. Held-out composition worsens mean image fidelity, and the full prerequisite audit admits no ScanNet++ policy rollout. Target-only native TEST comparisons show higher service success for the selection-and-retry bundle than fixed construction, but stricter verification reduces coverage and destination-scope reconstruction exposes further losses; privileged context and binding-specific predicates limit these conclusions. Prospective real-capture audits further expose the gap between reconstructing assets and obtaining usable simulators. All results retain failures, conditional denominators, and executable provenance.

I. INTRODUCTION

A robot entering a new room needs more than a visually plausible reconstruction. Its simulator needs movable objects, metric collision geometry, calibrated cameras, and observations that remain consistent with contact state. Recent real-to-sim systems advance interactive and compositional reconstruction [1], [2], [3], [4], [5], [6]. However, an unconditional cascade can propagate a locally unreliable output into an apparently complete simulator without revisiting the decision that produced it.

The relevant failures are heterogeneous. Different generators can complete different objects well; surface agreement need not imply stable support. A plausible asset can have incorrect scale, penetrate a support, or remain visually duplicated after insertion into a room. These failures motivate *agentic artifact construction*: maintaining explicit job state, comparing specialized tool outputs in a common frame, and using measurable evidence to decide the next action. Here, *agentic* denotes stateful feedback under predeclared rules. The controller does not require a language model,

and its benefit must be demonstrated separately from its implementation.

SimAnyRoom instantiates this process over room and object dependencies. Starting from posed images, a metric mesh, and a scene Gaussian representation, it discovers instances and assigns each a persistent identity. Generators propose assets; registration places their outputs in the observed metric frame; verification measures alignment, observation support, collision validity, and physical settling. An unsuccessful proposal can trigger a different bounded registration action. Every proposal, transform, decision reason, and terminal state is archived, including abstentions and failed attempts.

The same identity connects an accepted object’s collision body and visual asset. When the simulator moves the object, both follow its live state. Removing the original object Gaussians and completing the exposed background are separate construction obligations: synchronized motion alone does not prevent duplicate appearance. Static support and obstacle geometry define the contact environment. Optional appearance harmonization changes observations, while robot restoration and task-object preservation require their own tests.

Our central experiment separates mechanisms that are often conflated. A fixed single-tool constructor is compared with fixed-priority proposal diversity, evidence selection, bounded retry, and strict acceptance. The complete cohort shares planned jobs and initial proposals. Coverage uses all planned jobs; quality comparisons explicitly identify matched accepted objects. The results support evidence selection and registration retry as useful construction mechanisms. They also expose a substantial cost of abstention and a negative geometry–stability tradeoff under fixed proposal priority. Neither successful asset construction nor its acceptance-time physical probe certifies downstream manipulation.

A manifest-backed harness therefore treats manipulation as a separate test. It freezes policy, embodiment, cameras, control, resets, instructions, and rubrics while changing one treatment axis. A task interaction graph records which objects, supports, cameras, and robot frames are required for a query. Missing prerequisites remain visible instead of becoming unrecorded exclusions. Our contributions are:

- 1) A stateful real-to-sim construction process with explicit dependencies, common artifact contracts, and evidence-driven selection, retry, and abstention.
- 2) A persistent object representation linking metric geometry, collision, physical parameters, Gaussian appear-

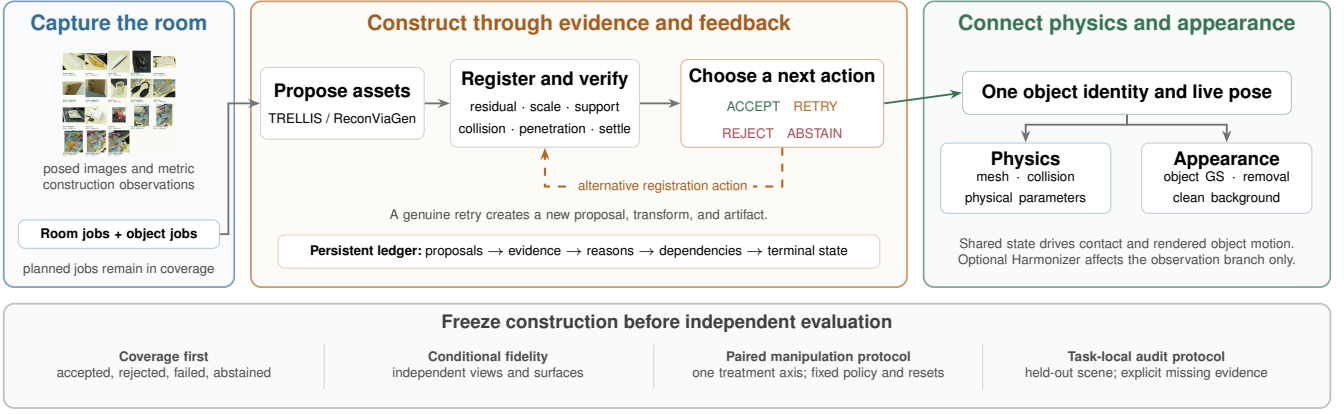


Fig. 1. **Evidence-driven artifact construction.** Room and object jobs invoke specialized tools, compare measurable construction outputs, and record accept, retry, reject, or abstain decisions. The evaluated retry changes registration with a signed source-up alternative; it does not invoke a new generator. The exported twin contract shares one object identity across physics and appearance. Independent evaluation consumes frozen outputs and never feeds reference geometry into the controller. TRELLIS.2 is a separate optional mesh-only adapter, outside the shared TRELLIS/ReconViaGen ablation pool. The capture image illustrates the input interface; downstream manipulation, Harmonizer, and task-local benefits remain unvalidated.

ance, and background removal, with a paired harness for testing the resulting simulator.

- 3) A complete controlled object-construction ablation across rooms, including common-support uncertainty and computation accounting, complemented by held-out appearance and real-capture audits that preserve failures and unsupported claims.

II. RELATED WORK

Automatic real-to-sim construction. RialTo [1] builds scene-specific twins with a human interface. MetaScenes [7] scales scene composition through annotations, ranking, and placement, while HoloScene [2], SimRecon [4], ReplicateAnyScene [5], SimFoundry [3], and RoboSnap [6] study increasingly automatic compositional recovery from images or video. These systems establish object factorization as a useful route from capture to simulation. SimAnyRoom focuses on the decision process around that factorization. It exposes scene and object jobs, permits multiple tools to propose artifacts, and uses common evidence to select or reject their outputs before they enter the simulator.

Neural rendering for simulation. Gaussian and neural scene representations provide photorealistic context but expose novel-view artifacts, missing shadows, and mismatches between inserted assets and reconstructed backgrounds. Harmonizer [8] uses a single-step, temporally conditioned enhancer to correct neural-rendering and insertion artifacts online. Our optional adapter places enhancement in the observation branch; its preservation and downstream gates remain open. Enhanced frames must preserve the robot and task object and must not reduce manipulation performance under a matched protocol.

Robot evaluation and reliability. SIMPLER [9] and PolaRiS [10] show that camera, rendering, and control matching are central to simulated policy evaluation. Real-is-Sim [11], RoboLab [12], and SureSim [13] further study simulator perturbations and reliability after an environment exists. SimAnyRoom couples automatic construction to a

reproducible paired harness. The harness makes the procedural contract executable, isolates declared scene, physics, and observation interventions, and retains unsupported cases in coverage. A hidden-oracle or primitive-proxy condition is never presented as real-world validation.

III. METHOD

A. Agentic Construction Graph

Given a captured room, SimAnyRoom instantiates a directed construction graph $\mathcal{B} = (\mathcal{J}, \mathcal{E})$. Nodes in $\mathcal{J}$ are explicit jobs. Room-level jobs recover metric scale, object discovery, background appearance, room collision, and robot alignment. Object-level jobs produce crops, candidate assets, metric registration, collision geometry, physical parameters, object Gaussians, removal sets, and verification evidence. Edges in $\mathcal{E}$ encode artifact dependencies and identify downstream products affected by a rejected registration. The evaluated controller applies feedback to object proposals and registration jobs; room-wide dependency invalidation remains an interface contract.

Each specialized tool k returns an artifact and an evidence record (a_{jk}, e_{jk}) for job j . Evidence includes provenance, dimensional and frame checks, construction-observation residuals, physical tests, and declared missing fields. Let $\mathcal{P}_j$ denote all proposals for job j . The scene controller applies a job-specific decision rule

$$\mathcal{P}_j = \{(a_{jk}, e_{jk})\}_k, \quad (1)$$

$$(a_j^*, u_j) = \mathcal{C}_j(\mathcal{P}_j, \mathcal{I}_j), \quad (2)$$

$$u_j \in \{\text{accept, retry, reject, abstain}\}, \quad (3)$$

where $\mathcal{I}_j$ denotes predeclared invariants. A retry requests another proposal or reruns a dependent tool with a declared alternative. Rejection discards one candidate. Abstention terminates an unsupported job and remains in coverage. A job ledger records proposal identities, parent dependencies, evidence, decision reasons, and terminal status. A retry must change an artifact or an executed action; reading the same

file again does not qualify. Missing evidence is explicit rather than a successful check. Evaluation annotations and oracle selection are unavailable to $\mathcal{C}_j$. Construction observations and registration targets are kept separate from evaluation references. This graph and feedback loop define the mechanism; their benefit requires the controlled ablation.

B. Persistent Object Construction

Given calibrated multi-view images $\{I_k, K_k, T_k\}$, a scene Gaussian field $\mathcal{G}$, and a metric mesh $\mathcal{M}$, open-vocabulary masks are lifted to $\mathcal{M}$ by first-hit ray casting, filtered for silhouette-depth spill, merged across views, and de-duplicated. For an exported twin, each object i has a persistent record

$$\mathcal{O}_i = (V_i, M_i, G_i, S_i, C_i, \theta_i, R_i), \quad (4)$$

where V_i is observed geometry, M_i and G_i are canonical mesh and Gaussian assets, $S_i \in \text{Sim}(3)$ places them in the room, C_i is collision geometry, θ_i contains physical parameters, and R_i is the Gaussian removal set. Physics, appearance, verification, and task grounding consume this record rather than recovering independent object identities. Controller acceptance alone does not establish completion of background removal or full-room export.

For each object, selected-view TRELLIS [14] and multi-view ReconViaGen [15] propose complete assets. The same symmetric clipped Chamfer objective registers every proposal into the observed metric frame. The evaluated controller orders proposals lexicographically: passing the evidence conjunction, fewer failed checks, lower clipped registration residual, smaller scale error, lower settle drift, then frozen tool and proposal order. A1–A4 share the same initial proposal pool. A3/A4 retry registration with alternative signed source-up axes only when all initial candidates fail. The retry creates a new proposal identity, registered artifact, and transform. A3 retains the best candidate with an unsupported flag if necessary; A4 abstains instead. The evaluated physical evidence uses one deterministic convex hull with fixed mass and contact parameters. Production asset export supports CoACD [16], but that decomposition and category priors are not the physical treatment measured in the controller table. The probe uses an isolated hull above a standard plane. Support overlap is the bottom-band to whole-hull footprint ratio; the initial gap and zero penetration are prescribed setup checks. They do not measure contact with reconstructed room supports.

An optional TRELLIS.2 [17] adapter supplies mesh and material assets through the same input identity and registration contract. Its separate pilot evaluates meshes only: it supplies no native Gaussian asset and remains outside the shared A0–A4 proposal pool.

The same identity drives appearance. Scene Gaussians are removed when they are close to the observed surface, close to the registered complete asset, or consistently projected inside the object’s masks across views. The last selector is opacity independent and captures diffuse Gaussians associated with transparent objects. Support-plane Gaussians complete the

hole and are refined against inpainted primary views, producing a clean background $\mathcal{G}^{\text{clean}}$. Moving body i moves M_i , C_i , and G_i together. Removal and completion errors remain appearance failures to evaluate.

C. Full-Room Physical and Neural Simulation

The exported twin is

$$\hat{E} = (\mathcal{G}^{\text{clean}}, \{\mathcal{O}_i\}, \mathcal{M}_{\text{room}}^{\text{coll}}, T_{\text{base} \leftarrow \text{scene}}). \quad (5)$$

Validated floor, support, and wall planes are represented by primitives, while remaining static geometry is convex decomposed. Room, object, and robot contacts share one collision system. This replaces private support shims, which support an object only at its initial footprint and cannot represent sliding, sweeping, or object–room contact after displacement.

At control step t , the renderer transforms every G_i by its MuJoCo body pose and composites the result over $\mathcal{G}^{\text{clean}}$. A MuJoCo renderer supplies robot RGB I_t^R and an exact robot mask M_t^R . The raw policy image is

$$I_t^{\text{raw}} = M_t^R \odot I_t^R + (1 - M_t^R) \odot \mathcal{R}(\mathcal{G}^{\text{clean}}, \{G_i, S_i(t)\}). \quad (6)$$

Physics, task scoring, and visual object placement therefore use the same live state.

D. Paired Robot Evaluation Harness

Let Γ denote the procedural contract: robot asset, policy checkpoint and preprocessing, policy cameras, action convention, controller, control rate, horizon, language instruction, task rubric, and random seed. Let ξ_n be a fully specified reset state generated before any condition is run. For treatment a , the harness records

$$y_{n,a} = \text{Rollout}(\hat{E}_a, \pi, \tau; \Gamma, \xi_n). \quad (7)$$

A comparison between a and b is admitted only when Γ and ξ_n are identical and their manifests differ solely along one declared treatment axis. The treatment axis may be construction policy, collision representation, or RGB observation.

For an eligible comparison, the harness generates and persists one reset bank. Every condition reuses that bank, independent of execution order or restart. For stochastic policies, the harness also binds sampling noise to the reset seed and action-chunk index. Warmup uses a separate random stream, so service startup, reconnects, and the order of preceding episodes cannot change a matched policy draw. Each control tick records the action, robot state, object poses, contacts, and task-stage predicates. Success, task failure, build failure, policy timeout, safety termination, environment crash, and enhancer failure remain distinct. An append-only ledger enables episode-level resume without resampling or duplicate rows. Coverage is therefore the fraction of planned episodes that produce a valid rollout, not the fraction remaining after failures are removed.

E. Observation Tool: Harmonization with Robot Restoration

Let $\mathcal{H}$ be Harmonizer and $h_{t,c}$ the temporal history for camera c . Full-frame enhancement produces $J_{t,c} = \mathcal{H}(I_{t,c}^{\text{raw}}; h_{t,c})$. Directly applying J can improve shadows and color consistency, but it can also alter the embodiment. We derive a blend $A_t^R \in [0, 1]$ from M_t^R . It is one on an eroded robot core, zero outside a dilated mask, and smoothly feathered only at the boundary. The robot-preserving observation is

$$I_{t,c}^C = A_t^R \odot I_{t,c}^{\text{raw}} + (1 - A_t^R) \odot J_{t,c}. \quad (8)$$

Robot-core pixels are copied exactly from the raw simulator. Histories are isolated by run, treatment, scene, task, reset, and camera, and reset between episodes. Benchmark runs never replace an enhancer failure with raw RGB.

F. Task Interaction Graph and Local Audit

Public task instructions and construction-grounded roles define

$$H_\tau = (R, C, O_m, O_t, S, \mathcal{O}_\tau), \quad (9)$$

containing the robot, policy cameras, manipulated object, target or receptacle, supports, and obstacles intersecting a robot-to-object approach proxy. Object-to-target transport-path coverage remains unvalidated. The graph expands by one support and visibility hop; other objects remain in physics and rendering but do not contribute to query risk.

For query τ , the audit collects

$$z_\tau = [z^{\text{vis}}, z^{\text{geom}}, z^{\text{support}}, z^{\text{robot}}] \quad (10)$$

from only the nodes in H_τ . Construction evidence includes mask and removal consistency, registration residual, candidate disagreement, scale, visibility, support overlap, penetration, settle response, reach margin, and robot-frame evidence. Publicly unresolved roles remain unresolved; language matching cannot replace them with benchmark object identities. Missing quantities are explicit and never imputed as a pass. The planned predictor uses a deliberately small calibrated model for invalid-query probability r_τ . For a fixed threshold δ ,

$$r_\tau \leq \delta \Rightarrow \text{ACCEPT}, \quad r_\tau > \delta \Rightarrow \text{ABSTAIN}. \quad (11)$$

The public feature table is sealed before reference-derived labels or the evaluation-only global appearance and geometry baselines are accessed. Neither labels nor evaluation baselines feed back into construction or its feature extraction. The predictive interface remains unvalidated; construction retries are evaluated separately.

IV. EVALUATION PROTOCOL

A. Cohorts and Construction Comparisons

We evaluate the entire validation cohort for novel views and semantics in ScanNet++ v2 (*nvs_sem_val*). Automatic discovery fixes the planned object jobs before generation. A0 uses TRELIS alone; A1–A4 share the initial TRELIS/ReconViaGen pool. A1 selects by fixed tool priority,

A2 by construction evidence, A3 adds a bounded signed-axis registration retry, and A4 applies strict acceptance with abstention. Retry creates a new proposal ID, transform, and artifact, without a new generator invocation. The same training observations and frozen evidence thresholds are used throughout; independent evaluation GT is unavailable to the controller. TRELIS.2 is evaluated separately as an optional mesh backend and is not added retrospectively to the shared ablation pool.

The broader input ladder compares GT and automatic discovery, scan and splat-fused meshes, and single-image metric depth. Its archived outputs have provenance limitations and are not used to establish input-regime equivalence. The fresh automatic splat-fused regime separately audits verified exports and an independent isolated-body drop test; it does not complete the input ladder. Room appearance uses a full predeclared common-view cohort, with cameras fixed before metric inspection and unavailable composites retained. The DROID cohort fixes workspaces and a chronological reference split before reconstruction: TRAIN robot kinematics determine temporal correspondence and metric alignment; disjoint reference kinematics evaluate the frozen fit. Because RGB can enter SfM, this is not an independent image-fidelity test.

B. Denominators, Metrics, and Computation

Coverage is accepted objects divided by all planned jobs, retaining missing proposals, build failures, rejections, and abstentions. Geometry uses symmetric Chamfer distance in centimeters and object-pooled F1@20 mm on independently matched accepted jobs. Unmatched geometry remains null; its objects remain in coverage. Registration targets and generator inputs are excluded from the independent evaluation reference. Catastrophic collapse is fixed as F1@20 < 0.1. Stable/tested counts use the controller’s frozen settle probe; this is construction evidence, not an independent physical experiment.

Attributed runtime combines original initial-generation process wall time, including failed and recovery processes, with canonical registration, physical probes, and retries. Shared computation can be charged to multiple policies. Capture, discovery, source Gaussian training, observation export, evaluation, queueing, and cache materialization are excluded. This quantity measures algorithm-phase cost rather than fleet elapsed or capture-to-simulator time. PSNR, SSIM, and LPIPS require pixel-aligned held-out views; object appearance requires independent masks. Those unmeasured comparisons remain distinct from matched object geometry.

C. Paired Effects and Independent Downstream Gates

All predeclared controller contrasts are retained. Coverage contrasts use every planned object; geometry uses common accepted objects matched to the same independent reference; stability uses common accepted, tested objects. Whole supported scenes are resampled with the frozen bootstrap protocol. The intervals are pointwise descriptive, without

multiplicity adjustment or seed replication. They do not convert conditional results into population-wide gains.

The manipulation harness changes exactly one construction, collision, or observation axis. Policy checkpoint, robot, cameras, action convention, controller, control rate, instruction, reset-state IDs, horizon, and rubric remain fixed. Persisted resets, tick records, unique episode IDs, and resumable ledgers expose failure and prevent resampling. Prerequisite checks precede rollouts; an ineligible task is not a measured policy failure. Eventual rollout metrics include coverage, success, grasp, lift, place, failure categories, and latency.

The task-support protocol seals construction features before validity labeling. Missing or failed object recovery, pose, scale, support, collision, robot frame, or required visibility makes a query invalid under fixed requirements. The planned low-capacity leave-one-scene-out comparison reports AUROC, AUPRC, Brier, Risk@80, and Risk@60 only when the label support admits estimation. Failed builds remain queries. Harmonizer requires aligned appearance, robot/target preservation, temporal-history isolation, and paired policy validation. Sim-real conclusions require genuine matched autonomous physical trials.

Each result is generated from hash-bound outputs. The controller’s independent completion audit verifies the frozen roster, metric shards, aggregate, uncertainty, and runtime replay; it does not rerun the matcher or supply a second geometry estimator. Publication records preserve original experiment freezes and exact source fields. Missing provenance, undeclared treatment drift, or silent raw-image fallback invalidates the corresponding claim.

V. RESULTS

Scope. The controller cohort is complete and independently audited. Other experiments retain the scope stated below. The publication imports hash-identified outputs from their original immutable runs; it is not yet a common submission freeze. Missing measurements are dashes, not zero.

A. Controlled Object Construction Across Rooms

The full cohort uses automatic discovery and TRAIN-only construction; independent GT is accessed only after controller outputs are frozen. Table I retains every policy’s accepted and planned counts, matched geometry denominator, collapse count, physical-probe support, retries, abstentions, and attributed runtime. Fig. 2 reports all fixed contrasts rather than selecting favorable metrics.

Diversity and evidence answer different questions. Fixed-priority proposal diversity improves coverage and matched geometry but reduces the construction-probe stable fraction. Thus, adding a second generator does not itself resolve conflicting evidence. Evidence selection improves matched geometry over fixed priority at unchanged coverage. The corresponding paired Chamfer and F1 intervals exclude zero in the improving direction on common accepted, independently matched objects. These comparisons hold the

initial pool fixed; the result concerns the controller’s choice among available proposals.

A different registration action can recover useful geometry. Bounded registration retries further improve matched geometry at unchanged coverage, with additional computation. A3 and A4 share these retry actions; re-reading an artifact is never counted as a retry. Attributed runtime includes original generation processes, failures and recoveries, plus canonical registration, physical probes, and retries; it is not capture-to-simulator or fleet elapsed time. The runtime increase accompanies the geometric benefit and remains visible in the same table.

Strict acceptance exposes a coverage tradeoff. A4 changes the accepted geometry subset; its conditional mean must not be read as a paired improvement over all planned jobs. Its larger conditional F1 and smaller Chamfer distance therefore do not establish unconditional superiority. Even the paired A4–A0 geometry contrast applies only to common accepted, matched objects, whose support is shown in the figure. A4’s perfect conditional stability is also selection conditioned: passing that same probe is required for acceptance. It does not independently validate contact dynamics or manipulation. Abstentions remain in coverage.

B. Export Yield and Independent Isolated Stability

The measured input regime retains every discovered object and reports verified exports before conditional isolated-body stability. Table II separates exported bodies from controller acceptance. The independent drop protocol tests the exported collision assets; it differs from the acceptance-time probe and does not certify full-room contacts. Room export rejections and empty scenes remain in the planned population. Other input regimes, exported-body held-out geometry, and complete runtime remain unmeasured.

C. Held-Out Room Appearance

The full planned room cohort retains unavailable composites and evaluates both measured methods on identical supported TEST views. Table III uses official views excluded from Gaussian training and a pinned AlexNet LPIPS backend. Raw renders exist for the entire planned roster; quality is restricted to the common composite support. Composite rendering has lower mean PSNR and SSIM and higher mean LPIPS than the raw Gaussian; this comparison provides no evidence of appearance improvement. These are descriptive means, without a paired-difference significance claim. Zero-object scenes retain unchanged backgrounds and are included in coverage, not counted as successful interactive simulators. The camera roster and construction eligibility were fixed before TEST metrics. The separate object-appearance and oracle-method roster remains unmeasured; the controller cohort evaluates geometry only on independently matched jobs. Harmonizer likewise has no admissible full visual comparison in this report.

The separate TRELIS.2 mesh-only pilot remains an optional backend check, without established geometry, ap-

TABLE I

COMPLETE AUTOMATIC-DISCOVERY CONTROLLER COHORT. COVERAGE RETAINS ALL PLANNED JOBS; GEOMETRY USES MATCHED ACCEPTED JOBS. A4 CHANGES THAT SUBSET. STABLE/TESTED IS THE ACCEPTANCE-TIME CONSTRUCTION PROBE. ATTRIBUTED RUNTIME INCLUDES INITIAL GENERATION, CANONICAL REGISTRATION, PROBES AND RETRIES, INCLUDING FAILED ATTEMPTS; IT IS NOT FLEET OR CAPTURE-TO-SIMULATOR TIME.

Policy	accepted/planned	coverage	geom. n	F1@20	CD [cm]	collapses	stable/tested	retries	abstain	attrib. min/scene
A0: fixed TRELIS	1640/1871	87.7%	517	0.301	11.767	126	471/1640	0	0	13.58
A1: fixed priority	1800/1871	96.2%	566	0.336	7.638	97	294/1797	0	0	35.68
A2: evidence selection	1800/1871	96.2%	566	0.348	7.196	99	659/1800	0	0	35.68
A3: bounded registration retry	1800/1871	96.2%	566	0.383	5.720	78	724/1800	1462	0	70.82
A4: retry + abstention	399/1871	21.3%	134	0.425	4.257	17	399/399	1462	1472	70.82

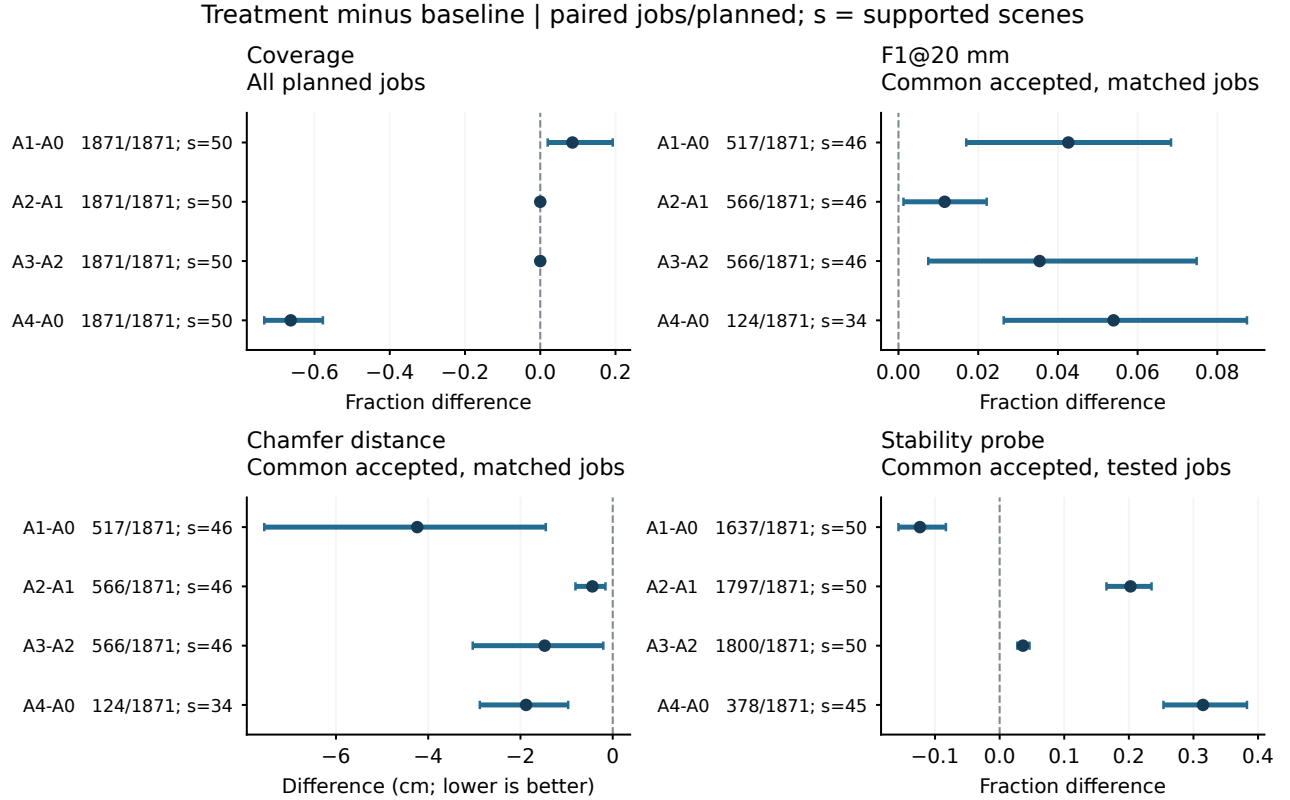


Fig. 2. All predeclared controller contrasts, with pointwise scene-cluster bootstrap intervals. Each label gives paired jobs/planned jobs and supported scenes. Coverage includes every planned job; geometry and stability use their stated common support. Geometry improvements from selection and retry coexist with a coverage cost from strict acceptance. Stability is the construction probe used by the controller, not independent physics validation.

TABLE II

FULL PLANNED INPUT LADDER, WITH ONE MEASURED EXPORT/DROP STAGE. EXPORT YIELD RETAINS ALL DISCOVERED INPUTS; STABLE/TESTED USES AN INDEPENDENT ISOLATED-BODY DROP PROTOCOL, DISTINCT FROM THE CONTROLLER'S ACCEPTANCE PROBE AND FULL-ROOM TASK QUALIFICATION. OTHER INPUT FAMILIES, EXPORTED-BODY HELD-OUT GEOMETRY AND COMPLETE RUNTIME REMAIN UNMEASURED. SUCCESSFUL EXPORTS DO NOT ESTABLISH COMPLETE INTERACTIVE SIMULATORS OR MANIPULATION.

Regime	input/scenes	exports	yield	F1@20	stable/tested	min/scene
GT segments + scan mesh	-/50	-	-	-	-/-	-
Auto discovery + scan mesh	-/50	-	-	-	-/-	-
GT segments + splat-fused mesh	-/50	-	-	-	-/-	-
Auto discovery + splat-fused mesh	1871/50	394	21.1%	-	361/394	-
Single RGB + metric depth	-/50	-	-	-	-/-	-

pearance or manipulation benefit; detailed results remain in the extended report.

D. Downstream Applicability

The ScanNet++ prerequisite audit admits no policy rollout: robot frames and policy cameras are unavailable. Its

TABLE III

FULL PLANNED HELD-OUT ROOM COHORT. QUALITY USES IDENTICAL COMMON AVAILABLE TEST VIEWS FOR RAW AND AUTOMATIC COMPOSITE ROWS; COMMON/PLANNED PRESERVES UNAVAILABLE COMPOSITES. RAW RENDERING IS AVAILABLE FOR EVERY PLANNED VIEW. ZERO-OBJECT SCENES WITH UNCHANGED BACKGROUNDS REMAIN INCLUDED AND ARE NOT SUCCESSFUL INTERACTIVE SIMULATOR BUILDS. GT-DISCOVERY AND HARMONIZER ROWS REMAIN UNMEASURED. MEANS ARE DESCRIPTIVE; NO PAIRED-DIFFERENCE SIGNIFICANCE IS CLAIMED.

Representation	scenes common/planned	views common/planned	PSNR	SSIM	LPIPS
Raw input Gaussian	40/50	320/400	21.582	0.8444	0.2993
Composite, GT discovery	0/50	0/400	–	–	–
Composite, automatic discovery	40/50	320/400	20.438	0.8318	0.3168
Harmonizer	0/50	0/400	–	–	–

TABLE IV

PROSPECTIVE DROID STAGES. EVAL./PLANNED PRECEDES HELD-OUT POSITION RMS (CM) AND MEDIAN ORIENTATION RESIDUAL (DEG). GS: COMPLETED (C) OR UNAVAILABLE (–); PREP.: PREPARED PROPOSALS, NOT ACCEPTED SIMULATOR ASSETS. ALL WORKSPACES AND FAILURES ARE RETAINED.

Workspace	CPU state	Eval./planned	cm	deg	GS / prep.
IPRL (1)	complete	16/47	1.82	2.80	C / 0
AUTOLAB	failed	0/16	–	–	– / –
TRI	failed	0/40	–	–	– / –
RAIL	complete	18/18	7.27	132.66	C / 0
RPL	complete	19/39	6091034.63	23.17	C / –
CLVR	failed	0/31	–	–	– / –
IRIS	failed	0/46	–	–	– / –
PENNPAL	complete	18/18	1.15	3.09	C / 0
REAL	failed	0/40	–	–	– / –
IPRL (2)	complete	46/46	0.48	1.24	C / 6

task-support labels contain only invalid queries, so LOSO prediction is not estimable. The extended report retains both complete audits; neither is evidence of successful manipulation or task-local prediction.

E. Prospective Real-Capture Alignment

The prospective DROID cohort reports independent kinematic alignment after TRAIN-only fitting; complete real-capture simulator construction remains unvalidated. All declared workspaces remain in the CPU-stage table; errors are conditional on the available held-out reference frames, and failed constructions retain missing measurements. RPL retains catastrophic held-out SfM camera-position outliers despite passing the frozen TRAIN translation gate. RAIL retains a large orientation discrepancy already present in TRAIN; this audit does not isolate calibration from SfM orientation error. Neither failure is clipped. A translation fit alone therefore cannot certify a usable robot-frame reconstruction. The continuation reports Gaussian training and prepared automatic proposals, retaining empty reconstruction and zero-discovery outcomes; it does not establish complete simulators. These stages use sealed TRAIN eligibility, without filtering by held-out error.

F. Native Sim–Recon–Sim: A Target-Only TEST Test

Across 48 frozen instances and 8 layouts, we reconstruct only the target from posed ideal RGB-D, using uniform materials and retaining native room/destination context. Policy,

TABLE V

NATIVE TARGET-ONLY TEST COMPARISON. COVERAGE RETAINS BUILD FAILURES AND ABSTENTIONS. DASHES DENOTE INCOMPLETE OUTCOMES; CONDITIONAL SUCCESS USES EXECUTED EPISODES.

Method	Exec./plan	Success/plan	Cond. SR
REF	480/480	381/480	79.4%
B0: fixed	310/480	79/480	25.5%
B3: agent	320/480	121/480	37.8%
B4: repair	120/480	65/480	54.2%
BM: fixed order	120/480	65/480	54.2%

robot, cameras, control, resets and horizons remain fixed. Table V counts build failures and abstentions against planned success; native success remains unmeasured for unexecuted units.

B0 is fixed TRELLIS; B3 adds shared TRELIS/ReconViaGen selection and bounded registration retry. B4 verifies and repairs context; BM uses the same action bank in fixed order.

The B3 bundle improves service success over B0 by 8.75 percentage points (paired hierarchical 95% interval [2.08, 16.25]); this does not isolate its components or control extra computation.

B4 raises conditional success but lowers planned success: its stricter acceptance loses coverage.

10 B0 prepolicy import failures were audited as construction failures; original geometry and native outcomes are unchanged.

Producer acceptance covers 32 assets versus 31 instances with native execution.

A separate fresh-control phase spans 24 instances and 4 layouts. Adding destination reconstruction changes L0-to-L1 service successes B3: 45/240 to 20/240; B4: 32/240 to 5/240. Native fixture articulations and goal handles remain.

Full-frame appearance has common support 12/48; retained background dominates.

Binding-specific predicates do not certify whole-mesh containment. These tests do not establish preservation, feedback benefit or complete room construction. Detailed protocols, intervals and failure audits remain in the extended report.

G. Remaining Evidence Boundaries

The archived input ladder retains all planned scenes but has incomplete original provenance. Splat-derived geometry is withheld where registration leaked scan geometry or artifact snapshots were inconsistent. Single-image frame selection used GT assistance. These historical outputs do not establish input-regime equivalence and are preserved in the machine-readable audit. Casual-phone construction remains incomplete. No matched autonomous physical trial cohort is available, so physical success, sim–real gap, and real-world predictivity are not reported. Demonstration replay and smoke outputs do not substitute for those experiments. Rejected builds, missing evidence, failed generation, and unavailable methods remain in the inventories; future policy timeout, safety stop, crash, and enhancer failure must remain terminal ledger rows.

VI. LIMITATIONS

The controller requires metric multi-view evidence and a declared manipulable- object prompt set. Rigid generated assets can have incorrect contact surfaces or mass distributions despite favorable geometry. The action space is limited to implemented tools and bounded retry rules. The full ablation establishes matched-object construction effects across rooms, not complete autonomous room simulator construction or policy transfer.

Conditional comparisons retain selection effects. Strict acceptance changes which objects are evaluated, and its own settle probe cannot independently validate physics. Scene-cluster intervals quantify variation over the observed cohort, without seed replication or multiplicity adjustment. Attributed computation excludes other room-construction stages. The incomplete input ladder, appearance matrix, and unavailable task/robot prerequisites limit system-level conclusions.

The native target-only track retains privileged room and destination geometry. Native task predicates depend on contact and body-origin bindings; they do not establish whole-object containment or independently corresponding frames. A completed TEST comparison does not establish noninferiority or preservation without a predeclared criterion. Separate scope and Gaussian diagnostics cannot establish complete room construction or correct robot occlusion.

A paired harness can freeze incorrect embodiment or calibration as faithfully as correct settings. Real physical trials remain necessary for sim–real claims. Harmonizer may alter task objects or add latency, and the task-local predictor cannot be evaluated without valid construction and reference support. These gaps are retained as failed or unrun claim gates.

VII. CONCLUSION

SimAnyRoom treats room-scale real-to-sim as evidence-driven artifact construction. Persistent object identities connect geometry, contact, and appearance, while explicit job state records selection, retry, and abstention. The complete controlled cohort shows that evidence selection and bounded

registration retry improve matched geometry at unchanged coverage, with additional computation. It also exposes the limits of proposal diversity and the coverage cost of strict acceptance. Constructing better object artifacts is a measurable step toward usable room simulators; downstream applicability and independent physical validation remain separate obligations.

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